

ALEJANDRO ARIAS ROJAS

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PROFESSIONAL SUMMARY

Business Informatics student (UCR, expected 2027) who builds production-grade business software, not classroom exercises. Shipped an offline-first ERP for a pharmaceutical laboratory with 372 automated tests and a native Windows desktop app, a POS and e-commerce platform live for a jewelry retailer for 9+ months, and a SaaS-oriented appointment platform. Full-stack across Python/Django, Node.js, React/Next.js, PostgreSQL/MySQL/SQL Server, Docker, AWS and Azure, with hands-on experience in backups, disaster recovery and production security hardening.

CORE COMPETENCIES

Backend Architecture • Database Design • Cloud Deployment (AWS / Azure) • Production Operations (Backups, Monitoring, Security) • Business Process Analysis • Technical Documentation

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL (T-SQL)

Backend: Django, Django REST Framework, Node.js, Express, Spring Boot

Frontend: React, Next.js, TypeScript, Tailwind CSS

Databases: PostgreSQL, MySQL, SQL Server, Redis

Cloud & DevOps: AWS, Azure, Docker, Docker Compose, Nginx, Linux, WSL2, systemd, Electron

Practices: Automated testing, role-based access control, backup/restore & rollback, offline-first architecture, security hardening (CIS benchmarks)

PROFESSIONAL EXPERIENCE

Independent Software Developer | Nov 2025 – Present

LICS — Offline ERP for Laboratorio de Inyección Castro Solís Jul 2026 – Present

- Designed and built an offline-first ERP covering inventory, purchasing, import costing, sales, customers, injector service tracking, physical counts and reporting for a real pharmaceutical laboratory.
- Architected a Dockerized stack (Django REST Framework, PostgreSQL 17, Next.js, Nginx) packaged as a native Windows desktop app (Electron + WSL2 + Docker Engine) for fully offline operation.
- Built an operational backup/restore/rollback system with automated systemd-timer backups, SHA-256 integrity verification and validated disaster recovery.
- Wrote 372 automated backend tests and full production-readiness documentation (architecture, security, deployment, troubleshooting).

Jewelry Management & POS Platform — Cuero&Perla Nov 2025 – Present

- Built and maintain a production monorepo (POS, admin backend, public storefront) for a working jewelry retailer, live for 9+ months across 800+ commits.
- Implemented point-of-sale, inventory with variants and composite products, accounts receivable, daily cash closing and returns.
- Built a public Next.js storefront with online ordering, cart, checkout and automated order-confirmation emails.
- Integrated thermal receipt printing (ESC/POS over WebUSB), Cloudinary image management and Web Push notifications; deployed on Railway and Vercel.

EsteticaPro — Appointment Management SaaS Jun 2026

- Built a scheduling platform for beauty salons with conflict-free booking and role-based access for admins, receptionists and estheticians.
- Implemented asynchronous reminder workflows with Celery, Celery Beat and Redis, including a WhatsApp notification integration.
- Designed as an API-first platform (Django REST Framework + JWT) intended for future multi-tenant commercialization.

ACADEMIC & TEAM PROJECTS — UNIVERSIDAD DE COSTA RICA

NexoAcadémico (solo, 2026) — Full-stack academic management platform (Next.js 16, Django 6, MySQL) with role-based workflows for admins, professors and students: enrollment, assignments, submissions and grading.

SIGAU — University Academic Management System (team of 3, 2026) — Enterprise database project on Azure (SQL Server 2025): CIS security hardening, Row-Level Security, Dynamic Data Masking, auditing and validated backup/restore on Windows Server.

Network & Security Infrastructure — IF5000 (team of 4, 2026) — Deployed and hardened a multi-service Linux server (Docker, Tailscale VPN, Suricata IDS, Fail2ban) with a real-time security alerting pipeline and Prometheus/Grafana monitoring.

TCP/IP Remote Administration System (team, 2025) — Client-server remote administration tool in Python using raw TCP sockets, multithreading and a custom GUI for remote monitoring, screen capture and mouse control.

EDUCATION

Universidad de Costa Rica (UCR) — Bachelor's Degree in Business Informatics | 2024 – 2027 (Expected)

Languages: Spanish (Native) | English (B2)